جامعة الملك عبدالله
للعلوم والتقنية
King Abdullah University of
Science and Technology

MACHINE LEARNING

– CS229 –

Final Exam

Total: 100 Points

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Student name: _____

Score: _____

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Problem 1: Sequential Gradient Descent in a Linear Regression Model

Statement. We have four one-dimensional data points:

$$(x_1, t_1) = (0,\ 0.1), \quad (x_2, t_2) = (1,\ 1.0), \quad (x_3, t_3) = (2,\ 1.9), \quad (x_4, t_4) = (3,\ 2.7).$$

We fit a linear model

$$y(x) \ = \ w_0 + w_1\, x,$$

where the sum-of-squares error function for all data is

$$E(\mathbf{w}) = \tfrac{1}{2} \sum_{n=1}^{4} \big[y(x_n) - t_n\big]^2.$$

The task is to perform *sequential* (online) gradient descent. We initialize $(w_0^{(0)}, w_1^{(0)}) = (0, 0)$, and we choose a fixed step size (learning rate) $\eta = 0.05$. Perform *one pass* over all 4 data points in the given order, showing each step.

(a) Write the partial derivatives $\frac{\partial E_n}{\partial w_0}$ and $\frac{\partial E_n}{\partial w_1}$ for a single data point (x_n, t_n), i.e. for $E_n = \frac{1}{2}\big(y(x_n) - t_n\big)^2$.

(b) Starting from $(0, 0)$, update (w_0, w_1) sequentially with step size $\eta = 0.05$ for the points in order: $(x_1, t_1), (x_2, t_2), (x_3, t_3), (x_4, t_4)$. Show the new (w_0, w_1) after each data point.

Grading:

- *(a) Formulation of derivatives – 6 pts*
- *(b) Sequential gradient descent – 12 pts*

Total = 18 points

Problem 2: Perceptron Learning in 2D

Statement. We have four two-dimensional data points, each labeled with a binary class $t_n \in \{-1, +1\}$.

$$(\mathbf{x}_1, t_1) = ((0,0), -1), \quad (\mathbf{x}_2, t_2) = ((1,0), -1), \quad (\mathbf{x}_3, t_3) = ((2,1), +1), \quad (\mathbf{x}_4, t_4) = ((3,2), +1).$$

We wish to use the *Perceptron* algorithm to find a separating hyperplane. The Perceptron decision is $\text{sgn}(w_0 + w_1 x_{n,1} + w_2 x_{n,2})$. We set the step size $\eta = 1$. A point is misclassified if

$$t_n\big(w_0 + w_1 x_{n,1} + w_2 x_{n,2}\big) \;\leq\; 0.$$

(a) Initialize $\mathbf{w}^{(0)} = (0, 0, 0)$ and present the data in the order $(\mathbf{x}_1, t_1), (\mathbf{x}_2, t_2), (\mathbf{x}_3, t_3), (\mathbf{x}_4, t_4)$, applying the Perceptron update rule whenever a point is misclassified.

(b) Perform second full pass through the data, showing how (w_0, w_1, w_2) evolves each time a misclassification occurs. Discuss whether the perceptron converges to a solution that correctly classifies all points.

Grading:

- *(a) Single pass logic – 10 pts*
- *(b) Second pass updates – 8 pts*

Total = 18 points

Problem 3: Backpropagation Computation in a Tiny Network

Statement. We have a tiny feedforward neural network for a single scalar input $x \in \mathbb{R}$ and one output y. The network has two hidden units, each with a logistic-sigmoid activation:

$$z_1 = \sigma(w_{1,1}x + b_1), \quad z_2 = \sigma(w_{2,1}x + b_2),$$

where $\sigma(z) = \frac{1}{1+e^{-a}}$. The output layer is linear:

$$y = w_{o1}\, z_1 + w_{o2}\, z_2 + b_{\text{out}}.$$

We have one training example $(x,\, t) = (2.0,\, 3.0)$. The error function is the sum-of-squares: $E = \frac{1}{2}\big[y - t\big]^2$. Assume an initial parameter setting:

$$w_{1,1} = 0.5, \quad b_1 = 0, \quad w_{2,1} = -0.2, \quad b_2 = 0.1, \quad w_{o1} = 1.0, \quad w_{o2} = 0.5, \quad b_{\text{out}} = 0.$$

(a) Perform a forward pass, computing $a_1,\, a_2,\, z_1,\, z_2$ and the final output y. Show numeric values.

(b) Compute all partial derivatives with respect to all weights and biases using the chain rule and write down all equations and then solve these also numerically.

(c) With a learning rate $\eta = 0.1$, update each parameter by one gradient descent step. Give the new parameter values.

Grading:

- *(a) Forward pass numeric – 6 pts*
- *(b) Partial derivatives (chain rule) – 8 pts*
- *(c) Single gradient update – 6 pts*

Total = 20 points

Problem 4: Gaussian Process Regression with an RBF Kernel

Statement. We have three 1D input–output data points:

$$(x_1, t_1) = (0,\ 0.4), \quad (x_2, t_2) = (2,\ 1.0), \quad (x_3, t_3) = (3,\ 1.8).$$

We place a Gaussian Process (GP) prior on the function $f(x)$ with zero mean and an RBF (radial basis function) kernel:

$$k(x, x') = \alpha \exp\left[-\frac{(x-x')^2}{2\,\sigma^2}\right],$$

where $\alpha = 1.0$ and $\sigma = 1.0$. There is *no noise* in the observations. We want the posterior distribution of $f_* \equiv f(x_*)$ at a new input $x_* = 1.5$.

(a) Construct the 3×3 kernel matrix $\mathbf{K}$ for the inputs $\{0, 2, 3\}$.

(b) Construct the vector $\mathbf{k}_*$ of size 3×1 with entries $k(x_*, x_n)$.

(c) Write down the general equations for posterior mean, covariance matrix and the predictive distribution and compute their values numerically at a new input $x_* = 1.5$..

Potentially useful inverses (you can use one of them, more of them or none):

$$\begin{pmatrix} 0 & 0.5 & 0.2 \\ 0.5 & 0 & 0.8 \\ 0.2 & 0.8 & 0 \end{pmatrix}^{-1} \approx \begin{pmatrix} -4 & 1 & 2.5 \\ 1 & 1.6244 & 0.625 \\ 2.5 & 0.625 & -1.5625 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0.1353 & 0.0111 \\ 0.1353 & 1 & 0.6065 \\ 0.0111 & 0.6065 & 1 \end{pmatrix}^{-1} \approx \begin{pmatrix} 1.0269 & -0.2088 & -0.1153 \\ -0.2088 & 1.6244 & -0.9828 \\ -0.1153 & -0.9828 & 1.5948 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0.0111 & 0.6065 \\ 0.0111 & 1 & 0.1353 \\ 0.6065 & 0.1353 & 1 \end{pmatrix}^{-1} \approx \begin{pmatrix} 1.5948 & 0.1153 & -0.9828 \\ 0.1153 & 1.0269 & -0.2088 \\ -0.9828 & -0.2088 & 1.6244 \end{pmatrix}$$

Grading:

- *(a) Kernel matrix (3×3) – 4 pts*
- *(b) Construct $\mathbf{k}_*$ (3×1) – 3 pts*
- *(c) Posterior mean & variance general + numeric – 8 pts*

Total = 15 points

Problem 5: Kernel-Based Regression from an Explicit Basis

Statement. We have four data points in 1D:

$$(x_1, t_1) = (0,\, 0.3), \quad (x_2, t_2) = (1,\, 1.2), \quad (x_3, t_3) = (2,\, 2.0), \quad (x_4, t_4) = (3,\, 2.8).$$

We define a set of *explicit* basis functions in $\mathbb{R}^3$:

$$\phi_1(x) = 1, \quad \phi_2(x) = \sqrt{2}\, x, \quad \phi_3(x) = x^2.$$

Then the feature map is $\boldsymbol{\phi}(x) = (\,\phi_1(x),\, \phi_2(x),\, \phi_3(x))$. We want to perform a *kernel-based* regression by first deriving the corresponding kernel function and then applying a kernel regression approach with regularization. Use a regularization parameter $\lambda = 0.5$.

(a) From the basis, derive the kernel function, write it out explicitly and simplify.

(b) Construct the kernel matrix.

(c) Perform kernel regression with regularization, express formula to learn vector $\mathbf{a}$, calculate those values and formulate the corresponding equation:

$$y_*(x_*) = \sum_{n=1}^{4} \alpha_n\, k(x_*,\, x_n).$$

$$\begin{pmatrix} 0.5 & 1 & 1 & 1 \\ 1 & 3.5 & 9 & 16 \\ 1 & 9 & 24.5 & 49 \\ 1 & 16 & 49 & 99.5 \end{pmatrix}^{-1} \approx \begin{pmatrix} -111.8462 & 31.6923 & 57.5385 & -32.3077 \\ 31.6923 & -9.3846 & -15.0769 & 8.6154 \\ 57.5385 & -15.0769 & -30.6154 & 16.9231 \\ -32.3077 & 8.6153 & 16.9231 & -9.3846 \end{pmatrix}$$

$$\begin{pmatrix} 4.5 & 9 & 16 \\ 9 & 25.5 & 49 \\ 16 & 49 & 100.5 \end{pmatrix}^{-1} \approx \begin{pmatrix} 0.9438 & -0.7031 & 0.1926 \\ -0.7031 & 1.1452 & -0.4464 \\ 0.1926 & -0.4464 & 0.1969 \end{pmatrix}$$

$$\begin{pmatrix} 1.5 & 1 & 1 & 1 \\ 1 & 4.5 & 9 & 16 \\ 1 & 9 & 25.5 & 49 \\ 1 & 16 & 49 & 100.5 \end{pmatrix}^{-1} \approx \begin{pmatrix} 0.8865 & -0.3841 & 0.0039 & 0.0504 \\ -0.3841 & 1.1102 & -0.7048 & 0.1707 \\ 0.0039 & -0.7048 & 1.1452 & -0.4462 \\ 0.0504 & 0.1707 & -0.4462 & 0.1998 \end{pmatrix}$$

Grading:

- *(a) Kernel function derivation – 5 pts*
- *(b) Kernel matrix numeric – 5 pts*
- *(c) Kernel ridge regression solution (matrix steps & final function) – 8 pts*

Total = 18 points

Problem 6: A Single ARD Update in an RVM Model

Statement. Consider a *Relevance Vector Machine* (RVM) regression setup in one dimension with two radial basis "kernels" (or basis functions). We have two data points (x_1, t_1) and (x_2, t_2), and the design matrix $\mathbf{\Phi}$ is 2×2:

$$\Phi_{n,i} \;=\; \exp\Big[-\tfrac{(x_n - c_i)^2}{2\sigma^2}\Big],$$

for two "centers" (c_1, c_2) in 1D. We assume a noise precision β, and each weight w_i has its own prior precision α_i (Automatic Relevance Determination).

Task: Suppose we already have a posterior over $\mathbf{w} = (w_1, w_2)$ with diagonal posterior covariance $\mathbf{S}$ and mean $\mathbf{m}$. Specifically, from some previous iteration:

$$\alpha_1 = \alpha_2 = 1, \quad \mathbf{m} = (m_1, m_2) = (0.2,\ 2.0), \quad \mathbf{S} = \mathrm{diag}(0.2,\ 0.04).$$

(a) Write down formula for γ_i and $\alpha_i^{(\text{new})}$ for $i = 1, 2$.

(b) Numerically compute γ_i and $\alpha_i^{(\text{new})}$ using necessary values as defined above.

(c) Conclude if either $\alpha_1^{(\text{new})}$ or $\alpha_2^{(\text{new})}$ appears like a potential candidate to prune the corresponding basis function.

Grading:

- *(a) Write update formulas – 3 pts*
- *(b) Numeric calculation for γ_i and $\alpha_i^{(new)}$ – 5 pts*
- *(c) Interpretation about pruning – 3 pts*

Total = 11 points